



ENGINEERING STUDENT CAREER COMPANION

Welcome back, Siddhartha! 🖐️

Every line of code you write and every problem you solve is bringing you closer to your dream software developer offer. Small, consistent steps build unstoppable momentum.

[Start Your Career Journey 🚀 →](#)

Today's Learning (2/5)

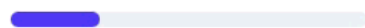


Daily Engineering Wisdom: "You don't have to be great to start, but you have to start to be great. Master one concept at a time."

ROADMAP

**25%**

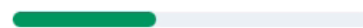
3 of 12 steps completed



TODAY'S TASKS

**2/5**

40% daily target done



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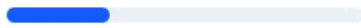
S

DSA PROBLEMS



10/35

29% problems solved

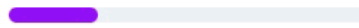


PROJECTS



1/4

3 in development

Your Current Milestone [View full 12-step roadmap >](#)Step 3 of 12 · [Programming](#) · 3-4 weeks

Python

Versatile language for backend development, scripting, data handling, and algorithmic problem-solving with expressive syntax.

In Progress

KEY SKILLS TO MASTER IN THIS STEP

- ☒ Built-in data structures (lists, tuples, dicts, sets)
- ☒ OOP implementation with classes and dunder meth...
- ☒ List, set & dictionary comprehensions
- ☐ Custom decorators and generator functions

Open Python Roadmap →



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Practice DSA

Solve curated Easy, Medium & Hard problems



Build Projects

Showcase real full-stack portfolio work



Career & Interview

Resume ATS tips, internship emails & Q&A

Today's Focus

[Manage all](#)

2 completed

3 remaining

~~Solve LeetCode #1: Two Sum (Optimal Hash Map approach)~~

DSA · 30 min

~~Build responsive navigation bar using Tailwind CSS Flexbox~~

Web Dev · 45 min



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Resume ATS tips, internship emails & Q&A

Today's Focus

[Manage all](#)

2 completed

3 remaining

~~Solve LeetCode #1: Two Sum (Optimal Hash Map approach)~~

DSA · 30 min

~~Build responsive navigation bar using Tailwind CSS Flexbox~~

Web Dev · 45 min



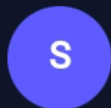
Review Git merge vs. rebase differences & commit conventions

Core CS · 20 min



Refactor user authentication routes with JWT verification

Project · 50 min

[+ Add or View All Tasks](#)

Siddhartha

Jain Institute of Technology Davangere
Software Development Engineer (SDE 1)

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Step-by-Step Curriculum

Software Developer Skills Roadmap

12 progressive milestones designed for engineering students: from web fundamentals to data structures, full-stack projects, and placement interview prep.

Roadmap Progress

25%

3 of 12 completed

5 in progress

All Steps (12)

In Progress (5)

Completed (3)



Step 1 · Foundation · 2-3 weeks



HTML & CSS

The visual and structural foundation of web applications....



Step 2 · Programming · 3-4 weeks



JavaScript

The core programming language of the modern web....



Step 3 · Programming · 3-4 weeks



Python



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3

Step 3 · Programming · 3-4 weeks



Python

OVERVIEW & OBJECTIVES

Versatile language for backend development, scripting, data handling, and algorithmic problem-solving with expressive syntax.

SET STATUS



In Progress



CORE COMPETENCIES (3/5)

60% mastered

~~Built-in data structures (lists, tuples, dicts, sets)~~~~OOP implementation with classes and dunder methods~~~~List, set & dictionary comprehensions~~

Custom decorators and generator functions



Exception handling with try/except/finally blocks

TOPICS TO MASTER



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TOPICS TO MASTER

- Pythonic Syntax, List & Dictionary Comprehensions
- Object-Oriented Programming (Classes, Inheritance, Dunder Methods)
- File Handling, Context Managers & Exception Handling
- Decorators, Generators & Functional Iteration
- Virtual Environments, Pip & Modular Code Organization

CURATED FREE RESOURCES & DOCUMENTATION

[Documentation](#)

Official Python Documentation & Tutorial

[Official & Free Link ↗](#)[Course](#)

CS50P - CS50 Introduction to Programming with Python

[Official & Free Link ↗](#)[Guide](#)

Automate the Boring Stuff with Python

[Official & Free Link ↗](#)

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MY STUDY NOTES & KEY TAKEAWAYS

Jot down formulas, code snippets, tricky interview traps, or revision notes for this topic...



Save Note

4

Step 4 · Programming · 3-4 weeks

**C/C++**

High-performance systems programming and the...

5

Step 5 · Core CS · 4-5 weeks

**Data Structures**

The structural foundation of efficient computer software....

6

Step 6 · Core CS · 4-6 weeks

**Algorithms**

Algorithmic problem-solving methodologies for technical...

7

Step 7 · Core CS · 2-3 weeks

**SQL**

Relational database architecture and structured querying...



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Step 8 · Foundation · 1-2 weeks

**Git & GitHub**

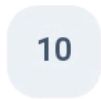
Version control workflows used by professional software...



Step 9 · Frontend · 3-4 weeks

**React**

Declarative component-based UI engineering. Master JS...



Step 10 · Backend · 4-5 weeks

**Backend Development**

Server-side engineering, RESTful API design, database...



Step 11 · Practical · 4-6 weeks

**Projects**

Translating knowledge into production-grade full-stack...



Step 12 · Career Prep · 4-6 weeks

**Placement & Interview
Preparation**

Comprehensive preparation for campus placement drive...



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Resume-Building Portfolio

Software Engineering Projects

Showcase full-stack engineering ability. Include project name, clear description, technology stack, status, GitHub source repository, and live deployed demo.

[+ Add Project](#)

All (4)

In Progress (1)

Completed (1)

Plai



Curated High-Impact Project Ideas for Engineering Students

Distributed Task Queue / Job Worker

Background worker handling asynchronous image processing or email queues using Redis and Node.js.

[+ Use this Project Idea](#)

Real-Time Markdown Collaborative Docs

Google Docs clone with operational transforms / CRDTs, live cursor tracking, and version history.

[+ Use this Project Idea](#)

Placement Preparation & Mock Assessment Engine

Platform for timed engineering assessments, automated



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Placement Preparation & Mock Assessment Engine

Platform for timed engineering assessments, automated code evaluation, and rank analytics.

[+ Use this Project Idea](#)

PROJECT NAME

In Progress



CampusConnect - Peer Tutoring & Study Material Exchange

DESCRIPTION

A full-stack collaborative platform for university students to share lecture notes, find study buddies by subject, and schedule peer tutoring sessions with real-time messaging.

KEY ACHIEVEMENTS / RESUME BULLETS

- Implemented real-time messaging with Socket.io and room-based authentication
- Created categorized document search with debounced text querying and indexing
- Engineered JWT-based role authorization for student learners and peer tutors

TECHNOLOGIES

React

Node.js

Express

MongoDB

Tailwind CSS

Socket.io



GitHub Code



Live Demo



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PROJECT NAME

Completed



AlgoVisualizer - Interactive Algorithm & Graph Simulator

DESCRIPTION

An interactive web tool that step-by-step visualizes pathfinding algorithms (Dijkstra, A*, BFS, DFS) and sorting routines (Quick Sort, Merge Sort) with customizable speeds and obstacle walls.

KEY ACHIEVEMENTS / RESUME BULLETS

- Visualized graph traversal animations with 60 FPS requestAnimationFrame rendering
- Provided step-by-step state inspection and pseudocode highlighter for learners
- Achieved 100% mobile-friendly responsive grid touch interaction

TECHNOLOGIES

TypeScript

React

HTML5 Canvas

Tailwind CSS



GitHub Code



Live Demo



PROJECT NAME

Planned



DevTrack - Software Engineering Internship Tracker



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PROJECT NAME

Planned



DevTrack - Software Engineering Internship Tracker

DESCRIPTION

A specialized job application tracker built for computer science students to manage application stages, cold email follow-up alerts, interview dates, and salary comparisons.

KEY ACHIEVEMENTS / RESUME BULLETS

- Kanban drag-and-drop board for Applied, Interview, Offer, and Rejected stages
- Automated email reminder triggers for pending recruiter follow-up windows
- Aggregated interview question log for company-specific debriefs

TECHNOLOGIES

Next.js

TypeScript

PostgreSQL

Prisma

Tailwind CSS



GitHub Code



Live Demo



PROJECT NAME

Planned



FastStore - High-Performance E-



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GitHub Code



Live Demo



PROJECT NAME

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FastStore - High-Performance E-Commerce Backend API

DESCRIPTION

RESTful microservice handling product catalogs, inventory reservations with Redis locks, order processing, and Stripe payment webhook integrations.

KEY ACHIEVEMENTS / RESUME BULLETS

- Designed normalized PostgreSQL relational schema handling 100k+ inventory SKUs
- Used Redis in-memory caching to reduce popular product query latency by 65%
- Integrated idempotency keys on payment routes to prevent double charging

TECHNOLOGIES

Node.js

Express

PostgreSQL

Redis

Docker



GitHub Code



+ Add Live Demo



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</> Technical Interview Foundation

Data Structures & Algorithms Practice

Solve curated problems across 10 essential categories: Arrays, Strings, Linked Lists, Stack, Queue, Hash Map, Trees, Graphs, Sorting, and Searching.

OVERALL SOLVED

10 / 35 (29%)

Easy Level

10/10 solved



Medium Level

0/17 solved



Hard Level

0/8 solved



Search problem title or pattern...

All

Easy

Medium

Hard

All

Solved

Unsolved

Bookmarked



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TOPIC CATEGORIES (10):

All 10/35

Arrays 2/4

Strings 1/3

Lir

~~Two Sum~~

Easy



Code

Arrays · Time: $O(n)$ ·Space: $O(n)$

Show Algorithm Intuition & Hint

~~Best Time to Buy and Sell Stock~~

Easy



Code

Arrays · Time: $O(n)$ ·Space: $O(1)$

Show Algorithm Intuition & Hint



Product of Array Except Self

Medium



Code

Arrays · Time: $O(n)$ ·Space: $O(1)$

Show Algorithm Intuition & Hint



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First Missing Positive

Hard

Arrays · Time: $O(n)$ ·Space: $O(1)$ 

Code

Show Algorithm Intuition & Hint



Valid Palindrome

Easy

Strings · Time: $O(n)$ ·Space: $O(1)$ 

Code

Show Algorithm Intuition & Hint



Longest Substring Without Repeating Characters

Medium

Strings · Time: $O(n)$ ·Space: $O(\min(n, m))$ 

Code

Show Algorithm Intuition & Hint



Minimum Window Substring

Hard

Strings · Time: $O(n + m)$ ·Space: $O(k)$ 

Code



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[Show Algorithm Intuition & Hint](#) 

Reverse Linked List



Code

Easy

Linked Lists · Time: $O(n)$ ·Space: $O(1)$ [Show Algorithm Intuition & Hint](#) 

Linked List Cycle

Easy



Code

Linked Lists · Time: $O(n)$ ·Space: $O(1)$ [Show Algorithm Intuition & Hint](#) 

Reorder List

Medium



Code

Linked Lists · Time: $O(n)$ ·Space: $O(1)$ [Show Algorithm Intuition & Hint](#) 

Merge k Sorted Lists



Code

Hard

Linked Lists ·

Time: $O(n \log k)$ ·

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Space: $O(k)$ [Show Algorithm Intuition & Hint](#) 

Valid Parentheses

Easy

[Code](#) Stack · Time: $O(n)$ ·Space: $O(n)$ [Show Algorithm Intuition & Hint](#) 

Daily Temperatures

Medium

[Code](#) Stack · Time: $O(n)$ ·Space: $O(n)$ [Show Algorithm Intuition & Hint](#) 

Largest Rectangle in Histogram

Hard

[Code](#) Stack · Time: $O(n)$ ·Space: $O(n)$ [Show Algorithm Intuition & Hint](#) 

Implement Queue using



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Implement Queue using
Stacks

Code

Easy

Queue ·

Time: $O(1)$ amortized ·Space: $O(n)$

Show Algorithm Intuition & Hint



Design Circular Queue



Code

Medium

Queue · Time: $O(1)$ ·Space: $O(k)$

Show Algorithm Intuition & Hint



Sliding Window Maximum



Code

Hard

Queue · Time: $O(n)$ ·Space: $O(k)$

Show Algorithm Intuition & Hint



Valid Anagram

Easy



Code

Hash Map · Time: $O(n)$ ·

Home



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~~Valid Anagram~~ Easy

Code

Hash Map · Time: $O(n)$ ·Space: $O(1)$

Show Algorithm Intuition & Hint



Group Anagrams



Code

Medium

Hash Map ·

Time: $O(n * k \log k)$ ·Space: $O(n * k)$

Show Algorithm Intuition & Hint



Longest Consecutive Sequence



Code

Medium

Hash Map · Time: $O(n)$ ·Space: $O(n)$

Show Algorithm Intuition & Hint

~~Maximum Depth of Binary Tree~~

Code

Easy

Trees · Time: $O(n)$ ·

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Space: $O(h)$ [? Show Algorithm Intuition & Hint](#) 

Validate Binary Search Tree

[Code](#)

Medium

Trees · Time: $O(n)$ ·Space: $O(h)$ [? Show Algorithm Intuition & Hint](#) 

Lowest Common Ancestor of a BST

[Code](#)

Medium

Trees · Time: $O(h)$ ·Space: $O(1)$ [? Show Algorithm Intuition & Hint](#) 

Binary Tree Maximum Path Sum

[Code](#)

Hard

Trees · Time: $O(n)$ ·Space: $O(h)$ [? Show Algorithm Intuition & Hint](#) 

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Number of Islands

Medium

Graphs · Time: $O(m * n)$ ·Space: $O(m * n)$ 

Code



Show Algorithm Intuition & Hint



Clone Graph

Medium

Graphs · Time: $O(V + E)$ ·Space: $O(V)$ 

Code



Show Algorithm Intuition & Hint



Course Schedule (Cycle Detection)

Medium

Graphs · Time: $O(V + E)$ ·Space: $O(V + E)$ 

Code



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Word Ladder

Hard

Graphs · Time: $O(m^2 * n)$ · Space: $O(m^2 * n)$ 

Code



Show Algorithm Intuition & Hint



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Sort Colors (Dutch National Flag)



Code

Medium

Sorting · Time: $O(n)$ · Space: $O(1)$

Show Algorithm Intuition & Hint



Merge Intervals

Medium



Code

Sorting · Time: $O(n \log n)$ · Space: $O(n)$

Show Algorithm Intuition & Hint



Kth Largest Element in an Array



Code

Medium

Sorting · Time: $O(n \log k)$ · Space: $O(k)$

Show Algorithm Intuition & Hint



Binary Search

Easy



Code

Searching · Time: $O(\log n)$ · Space: $O(1)$

Show Algorithm Intuition & Hint



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Search in Rotated Sorted Array



Code

Medium

Searching · Time: $O(\log n)$ · Space: $O(1)$ 

Show Algorithm Intuition & Hint



Find Minimum in Rotated Sorted Array



Code

Medium

Searching · Time: $O(\log n)$ · Space: $O(1)$ 

Show Algorithm Intuition & Hint



Median of Two Sorted Arrays



Code

Hard

Searching ·

Time: $O(\log(\min(m, n)))$ · Space: $O(1)$ 

Show Algorithm Intuition & Hint



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Placement & Hiring Accelerator

Engineering Career Development Hub

Everything you need to secure top software engineering internships and full-time job offers: ATS resume rules, cold outreach scripts, hiring workflows, and high-frequency interview Q&As.



Resume Preparation



Internship Preparation

Student Resume ATS Checklist

Applicant Tracking Systems (ATS) reject 70% of resumes before a human engineer ever reads them. Ensure your resume passes every check:



Clean single column layout (no multi-column tables or text boxes)



Contact info includes GitHub, LinkedIn, email, and location



Skills categorized into Languages, Frameworks, Tools, and Core CS



Project bullets follow Google XYZ formula: Accomplished [X] as measured by [Y], by doing [Z]



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☐ Includes at least 2 deployed full-stack or systems projects with live/code links

☒ ~~No arbitrary skill progress bars (e.g., 90% Python —ATS cannot read this)~~

☐ Strictly 1 page length for freshers and undergraduates

☐ Exported as clean, selectable PDF (not an image or flattened scan)

High-Impact Engineering Action Verbs

Replace passive verbs like "Worked on" or "Helped with" with decisive, impactful technical terms:

Building & Developing:

Architected Implemented Engineered
Constructed Formulated Deployed

Optimizing & Scaling:

Refactored Streamlined Accelerated
Reduced latency Optimized Automated

Testing & Security:

Validated Integrated Debugged
Standardized Secured Benchmarked



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The Google XYZ Resume Formula:

"Accomplished [X], as measured by [Y], by doing [Z]"

Example: "Reduced database query latency by 42% (Y) by adding multi-column B-Tree indexing in PostgreSQL (Z) across 50,000 student records (X)."

Ready-to-Use Student Developer Resume Template

Based on Jake's Resume (Overleaf / LaTeX standard used by top tech applicants).

[Copy Template](#)**SIDDHARTHA KUMAR**

Email: siddhartha@example.com | Phone: +91 98765 43210

GitHub: github.com/siddharthadev |

LinkedIn: linkedin.com/in/siddhartha-engineer | Portfolio:

siddharthadev.com

EDUCATION

Jain Institute of Technology Davangere

B.Tech in Computer Science and

Engineering | CGPA: 8.8 / 10.0 |

Expected Graduation: 2027

TECHNICAL SKILLS

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Based on Jake's Resume (Overleaf / LaTeX standard used by top tech applicants).

 Copy Template

SIDDHARTHA KUMAR

Email: siddhartha@example.com | Phone:
+91 98765 43210

GitHub: github.com/siddharthadev |
LinkedIn: linkedin.com/in/siddhartha-
engineer | Portfolio:
siddharthadev.com

EDUCATION

Jain Institute of Technology Davangere
B.Tech in Computer Science and
Engineering | CGPA: 8.8 / 10.0 |
Expected Graduation: 2027

TECHNICAL SKILLS

- Languages: C++, JavaScript, TypeScript, Python, SQL
- Frameworks & Libraries: React, Node.js, Express, Tailwind CSS,

